

An Exploratory Study of Hypergraph Neural Networks for Drug–Drug Interaction Modeling

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Abstract

Most drug–drug interaction (DDI) prediction pipelines represent interactions as pairwise edges in a graph. This assumption is convenient but may overlook higher-order structure when multiple drugs jointly shape an interaction event. We present an exploratory study that reframes DDI data as a hypergraph, where each hyperedge can connect more than two drugs, and compare a hypergraph neural network (HGNN) against graph baselines (GCN, GAT, GraphSAGE) and an MLP baseline. On a synthetic demo benchmark used for pipeline validation, we observe model-dependent differences across ROC-AUC, PR-AUC, and F1. We emphasize that these synthetic results are *not* pharmacological evidence and should be interpreted only as a validation of the end-to-end experimental workflow.

1 Introduction

Drug combinations can produce emergent effects that are not always reducible to independent pairwise relations. Standard graph neural network pipelines for DDI prediction typically represent interactions as binary edges, which simplifies modeling but can discard higher-order context. Hypergraphs provide a natural alternative by allowing one interaction event to link multiple drugs directly [1].

This report documents an exploratory implementation in the **hypergraph-ddi** project. The objective is not to claim clinical performance, but to evaluate how representation choices (graph vs. hypergraph) influence downstream behavior under a controlled training and evaluation pipeline.

The accompanying software artifact is available as hypergraph-ddi (Version 0.1.0) [5].

1.1 Motivating Example: Why Hypergraphs

Consider a single interaction event involving three drugs, A , B , and C . A pairwise graph encodes this as edges (A, B) , (A, C) , and (B, C) , but these edges do not preserve that the three drugs were co-administered in one event. For an illustrative clinical analogue, one may consider antithrombotic triple therapy (warfarin + aspirin + clopidogrel), where risk reflects the combined regimen rather than any single pair in isolation; major cardiology guidance highlights increased bleeding concern with such triple combinations and recommends limiting exposure when possible [3]. A hypergraph represents the same event as one hyperedge $\{A, B, C\}$, which retains event-level context and can express higher-order synergy patterns in an illustrative way.

Formally, representing an event as a hyperedge $e = \{A, B, C\}$ preserves joint incidence information that is not uniquely recoverable from the unordered set of pairwise edges alone.

2 Method

2.1 Problem Formulation

Let $\mathcal{V}$ denote drug nodes and $\mathcal{E}$ denote interaction events. In the graph view, each event is decomposed into pairwise edges. In the hypergraph view, each event is represented as a hyperedge $e \subseteq \mathcal{V}$ with cardinality $|e| \geq 2$.

The downstream task remains binary pairwise DDI prediction for comparability with common baselines and metrics.

2.2 Models

We evaluate five models:

- **HGNN**: higher-order message passing through node–hyperedge incidence structure [1].
- **GCN**: graph convolution baseline [4].
- **GAT**: graph attention baseline [7].
- **GraphSAGE**: neighborhood aggregation baseline [2].
- **MLP**: non-graph reference model over node features.

2.3 Implementation Notes

The repository applies a unified training/evaluation stack for all models, including shared preprocessing, split handling, optimization loops, and metric computation. Hypergraph and graph structures are derived from training interactions only to reduce structural leakage into evaluation.

3 Experimental Setup

3.1 Data Scope

This report focuses on the synthetic demo benchmark bundled with the project as a reproducible smoke-test environment. The synthetic setting verifies that preprocessing, training, metric logging, and multi-seed aggregation operate correctly.

The synthetic benchmark is configured as shown in Table 1. This setup is used for pipeline validation only and is not intended to support domain-level efficacy claims.

Table 1: Synthetic dataset configuration used in this study.

Setting	Value
Drug nodes	50
Hyperedges	80
Max hyperedge size	4
Feature dimension	64
Split	70/15/15
Negative sampling	1:1

Real-world sources (e.g., DrugBank [8] and TWOSIDES [6]) are supported by configuration but are not benchmarked in this report.

3.2 Protocol

All models are run with five seeds (42–46) under a common CPU-based pipeline. We report mean $\pm$ standard deviation for ROC-AUC, PR-AUC, and F1.

4 Results

Table 2 summarizes synthetic benchmark outcomes.

The repeated F1 values across several models are plausible under this setup and should not be over-interpreted. Likely causes include a fixed decision threshold of 0.5, small test-set size, limited test diversity, and score distributions concentrated near the decision boundary. For this reason, ROC-AUC and PR-AUC are emphasized as threshold-independent metrics; future work should also report confusion matrices or positive prediction rates to clarify threshold effects.

Table 2: Synthetic benchmark results (mean $\pm$ std over 5 seeds).

Model	ROC-AUC	PR-AUC	F1
HGNN	0.4893 ± 0.1034	0.7168 ± 0.0750	0.8235 ± 0.0000
GCN	0.5693 ± 0.0991	0.7549 ± 0.0722	0.3283 ± 0.4496
GAT	0.3824 ± 0.0850	0.6477 ± 0.0478	0.8235 ± 0.0000
GraphSAGE	0.4923 ± 0.0704	0.6909 ± 0.0419	0.8235 ± 0.0000
MLP	0.4970 ± 0.0998	0.7069 ± 0.0503	0.8235 ± 0.0000

5 Discussion

These synthetic outputs indicate that the pipeline can train and evaluate heterogeneous model families under a unified protocol. Variation across metrics suggests representation and inductive bias can change model behavior, even in controlled toy settings.

Importantly, HGNN does *not* outperform GCN on this synthetic benchmark: GCN attains the highest ROC-AUC and PR-AUC in Table 2. Therefore, the observed differences are interpreted as representation-behavior effects under a toy protocol, not as evidence of HGNN superiority.

However, this table should not be interpreted as evidence that any architecture is superior for biomedical deployment. The synthetic environment is intentionally simplified and is intended for sanity checking the workflow. Given the small synthetic benchmark and five-seed protocol, differences between models should be interpreted descriptively rather than as statistically significant performance gaps.

6 Limitations

- **Synthetic-only evidence:** results validate engineering flow, not pharmacological utility.
- **Feature realism:** default node features in demo settings are simplified placeholders.
- **Task reduction:** final scoring remains pairwise, even when training uses higher-order structure.
- **External validity:** no clinical, mechanistic, or prospective validation is included.

7 Next Steps

- Run the same protocol on curated real datasets with documented licenses and preprocessing.
- Add stronger drug representations (e.g., molecular descriptors, multimodal features).
- Evaluate calibration, subgroup behavior, and robustness under distribution shifts.
- Study objectives that directly model higher-order outcomes beyond pairwise reduction.

8 Conclusion

This project provides an exploratory, reproducible framework for testing hypergraph-based DDI modeling against common graph and non-graph baselines. The current synthetic benchmark confirms pipeline integrity and supports method development, but it should be treated strictly as pre-benchmark validation prior to real-data scientific claims.

References

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